Contents

1 Introduction

The Portuguese real estate market is one of the hottest and most popular real estate markets in Europe. For example, the median house price per square meter increased to 1,819 Euros in the third quarter of 2024, which is a 10.8% increase compared to the same period the year before. To illustrate some extreme cases, the region of Alto Alentejo saw the largest year-on-year increase over the same period: 26.3%! (Statistics Portugal, 2025)

More specifically, house prices were increasing 13 of the 24 most populated municipalities (more than 100,000 residents). Of this subgroup of populous municipalities, Maia saw the highest increase (13.8 percentage point year-on-year), while Porto even saw a rather steep decline (10.7 percentage point year-on-year). Lastly, the capital of Lisbon saw a 1.2 percent point increase. (Statistics Portugal, 2025)

But what drives these prices the way they move? Is it the increase in foreign investment by rich investors and people who want to spend their retirement in the sunny Algarve? Is it the people who buy real estate to profit of the important tourism sector in Portugal? Is the government slacking with bad policy? Or a combination of the above?

In this report, the aim is to come up with an explanation of the Portuguese housing market dynamics, and to subsequently formulate a strategy to regulate and possible reduce housing prices. In doing so, a spatial perspective will be adopted, meaning space, locational and geographical factors will be included in this analysis. Concretely, this means that Portugal will not be looked at as a single unit that is subject to analysis, but rather the sub-regions and municipalities and their spatial dependencies and relations that are under inspection. By using spatial descriptive statistics dimensionality reductions methods such as principal component analysis, underlying price drivers and important determining factors will be deducted and will ultimately lead to a policy recommendation to the Portuguese government.

2 Data description

The data set contains 38 variables, with economic, demographic, geographic and property natures. In this research, the aim is to make a distinction between demand-driving variables and supply-driving variables. Therefore, the variables in the data set are being split accordingly. In Table 1, the demand driving variables are displayed, and in Table 2, the supply driving variables are displayed as well as the dependent variables (i.e., house prices).

The variables in Table 1 are grouped as demand drivers because they either contain demographic variables, such as population (and variables derived from it, such as population density, % or population with a degree, etc.) or contain economic variables such as purchasing power, unemployed population, IMI/IMT tax revenue). These are all variables that direct or indirectly impact the quantity of demand for housing and real estate. Lastly, there are also some remaining variables that might influence demand because they determine how attractive a place is (crimes recorded, number of hospitals, schools).

Next, in Table 2, the supply variables are grouped as they are because it concerns existing real estate, such as the variables new houses, touristic/classical accommodations, entities for sale and new/total buildings. These variables describe the physical(potential) market supply that is available in Portugal. On the other hand, there are variables that describe existing home owners, who might sell/supply their real estate in the future. Lastly, there are ancillary variables that might somehow influence how effective supply is regulated such as the winner of municipal elections, who might determine how much houses are allowed to be build.

Lastly, there are two candidate dependent variables that reflect house prices (in slightly different ways).

	Drivers of Demand	
Classical households	Active population	Museums
Purchasing power	Crimes recorded	Environmental NGOs
Net population variation	Unemployed population	Electricity consump. / inhabitant
Net migration	% resident with Portuguese nationality	Vacation rentals
Population > 15 years	Foreign guests in tourist accom.	% pop. > 15 that hold a degree
Population that enters/leaves	Number of students	% population > 65
Population density	Hospitals	IMI tax revenue
Total population	Schools	IMT tax revenue

Table 1: Demand drivers

Supply		**Dependent**
% of owners with charges	Classical accommodations	Median bank valuation/m2
% of owners with charges > 400 EUR p/m	Winner municipal elections	Av. value of buildings transacted
New houses	Buildings: New constructions	
Touristic accommodations	Buildings: total	
% houses served by drainage systems	No. of entities for sale	

Table 2: Supply drivers and dependent variables

3 Methodology

In this study, we aim to model housing prices in Portugal and propose a strategy to reduce them. Given the high dimensionality of our dataset, which contains numerous variables potentially influencing housing prices, our first step involves reducing complexity while preserving interpretability. To achieve this, we categorize all features into two conceptual groups: demand-side factors (such as population density, average income, and employment rate) and supply-side factors (including housing stock, construction activity, and land availability). This separation allows us to capture the distinct dynamics driving demand and supply in the housing market.

We apply Principal Component Analysis (PCA) separately to the demand-side and supply-side variables to reduce dimensionality and extract the main components that explain most of the variance in each group. This results in two sets of principal components—one summarizing demand-related variation and the other summarizing supply-related variation. These principal components serve as the primary explanatory variables in our subsequent analysis.

To explore how these components relate to housing prices spatially, we compute local Moran's I statistic for each principal component and for housing prices themselves. This allows us to detect and quantify spatial autocorrelation, providing insights into the degree to which similar values cluster geographically. Understanding these spatial patterns helps reveal how local dynamics contribute to price differences and identifies regions where targeted interventions may be most effective.

Following the spatial analysis, we fit an Ordinary Least Squares (OLS) regression model, using the housing price variable as the dependent variable and the extracted demand and supply principal components as independent variables. The OLS model offers a baseline estimation of how underlying economic and structural factors influence prices.

However, residual analysis of the OLS model indicates the presence of spatial autocorrelation in the errors. This suggests that unobserved spatial effects remain. To address this issue, we implement spatial regression models.

By combining dimensionality reduction, spatial correlation analysis, and spatial econometric modeling, our methodology provides a comprehensive framework for understanding the determinants of housing prices in Portugal.

4 Descriptive statistics

The *median bank valuation per square meter* is provided in figure 1. It can be observed that the median bank valuation per square meter is the highest in the Algarve, at the coast, Lisbon and Porto. Next, there is information on the average value of buildings transacted below in figure 2. The distribution of the prices across Portugal is about the same as for the median bank valuation, only the data from the average value of buildings transacted is more outdated (from 2019).

As can be seen, there is a relatively large portion of the municipalities for which there is no data (indicated by white). More specifically there are 61 out of the 278 municipalities for which a there is no value. However, when looking at the number of households that is actually missing, this is only 3.8% of total households.

Lastly, the population density is provided in figure 2 to provide some knowledge about the distribution of inhabitants across the country, as well as a figure indicating where most new real estate is build (figure 4).

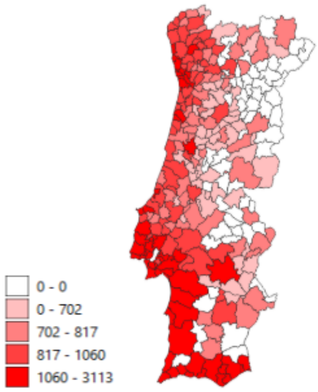

Figure 1: Median bank valuation (Euros)

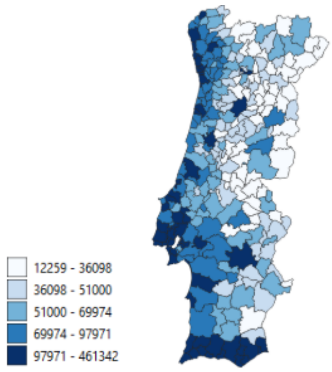

Figure 2: Average value of buildings transacted (Euros)

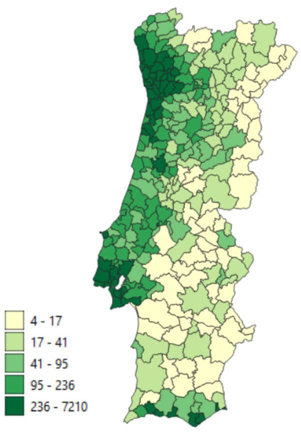

Figure 3: Popuation density

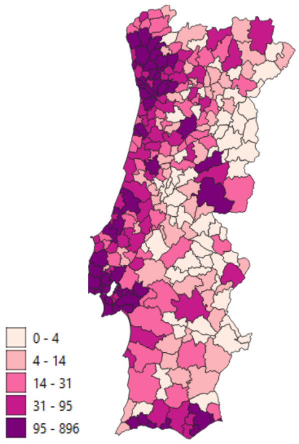

Figure 4: New houses

Figure 5: Mapping of the principal components

5 Principal component analysis

5.1 Defining principal components

For spatial data analysis, a principal component analysis will be used with the main argument of dimensionality reduction. In this research, a large variety of variables are included. Because the aim is to detect underlying patterns and drivers, the aim is to reduce the regressor space while retaining important information.

In table 3, the results of the *demand* principal component analysis are visible. Note that only the principal components with an eigenvalue above one are reported as these are in accordance with the Kaiser criterion.

In table 4, the results of the supply principal component analysis are visible. Again centroids are included and only principal components with an eigenvalue above one are reported.

	PC1	PC2	PC3	PC4	PC5
Proportion of variance	0.550848	0.105487	0.087596	0.052083	0.03873
Cumulative proportion	0.550848	0.656336	0.743931	0.796014	0.834744
Eigenvalues	15.4238	2.95365	2.45269	1.45832	1.08445

Table 3: Demand PCA

	PC1	PC2	PC3	PC4
Proportion of variance	0.487379	0.135589	0.129356	0.095067
Cumulative proportion	0.487379	0.622968	0.752324	0.847391
Eigenvalues	5.36117	1.49148	1.42292	1.04574

Table 4: Supply PCA

As can be seen in table 3, the total amount of variance explained by the significant demand side principal components is 83.4% . For the supply side principal components, table 4 indicates that the total amount of variance explained is 84.7%. For both demand and supply, the objective of variable space reduction is achieved while retaining important information that the individual variables contain.

When all principal components are analyzed with respect to which variables have a strong correlation with each respective principal component, the loadings and the distribution in space, we identify the following principal component 'names' are obtained (we use 'D' for the demand PC's): D1 is the *population* principal component, as it is highly correlated with the number of households, different population variables, number of students and IMI tax revenues, among others, which are all dependent on the size of the population. Next, D2 seems to be a *internal migration* principal component, as it is correlation with net population variation and net migration. D3 seems to be a *tourism* related principal component, as it positively correlated with the percentage of Portuguese nationals and the energy consumption. The reasoning behind this slightly odd name is that when the percentage of Portuguese nationals is higher in a region, there is less tourism as well as when the energy consumption is higher in a municipality, there is less tourism as tourism is typically not as strong all year round so there is less energy consumption in total. D4 is a *migration* principal component, as it only correlated to net migration. For the analysis, this is the migration PC. Lastly, D5 is only correlated with the municipal tax rate, so D5 is the *municapality tax* component.

With respect to the **supply** principal components, S1 can be interpreted as the *accommodation supply* principal component, as it is correlated with classical and new houses. The second principal component can be interpreted as the *North-South difference* principal component, as it is correlated with the Y or longitude centroid. S2 also partly captures touristic supply in the South. Next, S3 can be named the *touristic accommodation* supply. Lastly, S4 is a principal principal component that captures the information about supply that is not captured by the first three components. Moreover, the principal component is particulary strong in the *east* of Portugal.

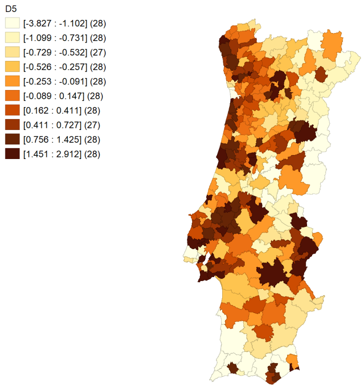

Figure 11: D5

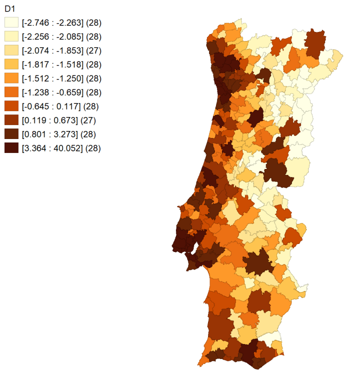

Figure 6: D1

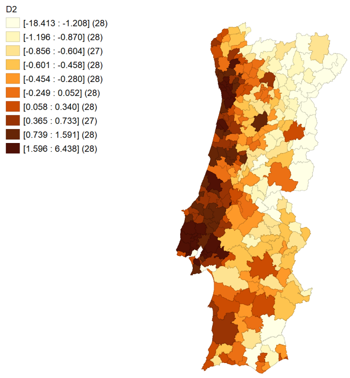

Figure 7: D2

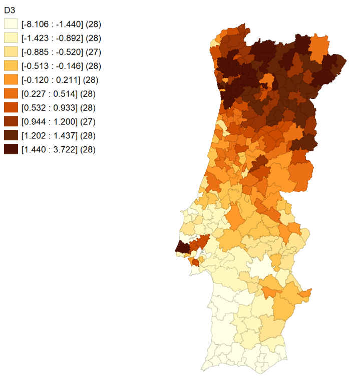

Figure 8: D3

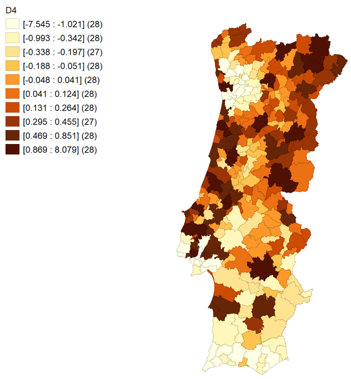

Figure 9: D4

Figure 10: Mapping of the demand principal components

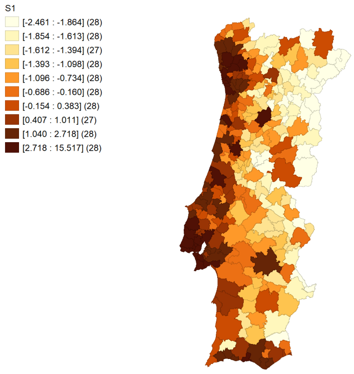

Figure 12: S1

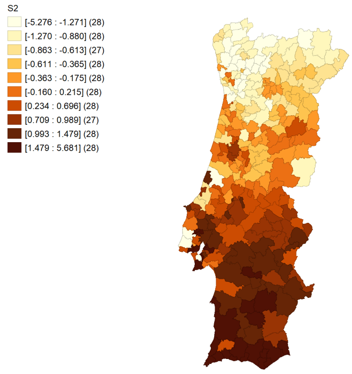

Figure 13: S2

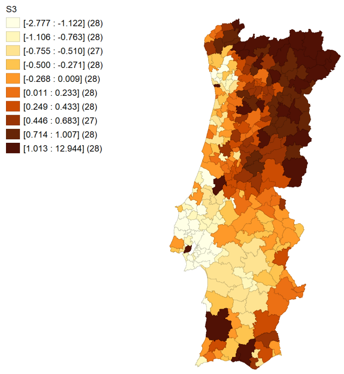

Figure 14: S3

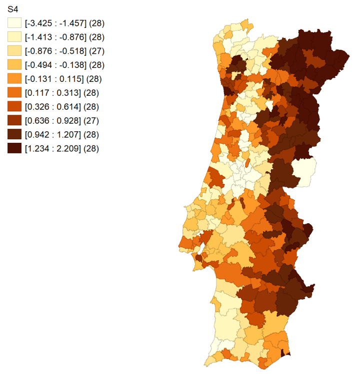

Figure 15: S4

Figure 16: Mapping of the supply principal components

6 Spatial autocorrelation

After doing principal component analysis and running regression on the dependent variables, spatial autocorrelation will be checked. Spatial autocorrelation is a relevant concern because we want to see if variables are clustered or not (i.e. dispersed) over Portugal. To check for this, we use *Moran's I*, which is a statistical test for this phenomenon. The test statistic takes values between -1 and 1. Here, a negative test statistic means spatial autocorrelation, meaning dispersion of similar values. Next, a positive test statistic means positive autocorrelation, meaning clustering of similar values. Here, the magnitude of the test statistic indicates how strong the dispersion and clustering are, respectively. Lastly, a test statistic of 0 means that there is no spatial autocorrelation, which means spatial independence (no clear clustering or dispersion). In this section we discuss how our most important principal components are spatially correlated with real estate prices on a local level.

An analysis of the Local Moran's I statistics for both price variables in relation to the first demographic variable (D1: Population) reveals a predominance of non-significant spatial associations across much of the study area. However, distinct spatial clusters emerge in certain regions. Notably, the Lisbon, Porto, and Algarve regions exhibit statistically significant high-high clusters, indicating areas where high price levels coincide with high population densities. Conversely, in the eastern part of Portugal, low-low clusters are evident, reflecting a spatial concentration of low prices alongside low population figures. These patterns suggest a spatially uneven distribution of price dynamics that align with regional demographic concentrations.

The spatial relationship between internal migration (D2) and housing prices reveals distinct regional patterns across Portugal. Areas experiencing net in-migration, such as Lisbon, Porto, and the Algarve, are characterized by significantly high housing prices, suggesting that population influx contributes to increased demand and price escalation in these urban and coastal regions. In contrast, eastern regions of the country, which are marked by net out-migration, tend to exhibit low housing prices, reflecting depopulation pressures and diminished housing demand. Between these two dynamics, a broad swath of central Portugal displays largely non-significant spatial associations.

The spatial association between tourism activity (D3) and housing prices reveals important regional patterns, shaped by the inverse nature of the tourism principal component (PC), where lower values correspond to higher levels of touristic activity. High tourism intensity and elevated housing prices are particularly evident in the Algarve, forming high-high clusters that suggest a strong connection between tourism-driven demand and price escalation in this coastal region. In contrast, eastern Portugal displays low-low clusters, indicative of limited tourism activity and correspondingly lower housing prices. Interestingly, Lisbon and Porto—traditionally recognized as major urban tourist destinations—do not exhibit as strong a relationship between D3 and housing prices as one might expect. This discrepancy may point to a limitation or imperfection in the construction or sensitivity of the tourism principal component, which may underrepresent tourism intensity in these metropolitan areas despite their well-known touristic appeal.

An examination of the spatial association between accommodation supply (S1) and housing prices further reinforces the regional disparities observed in Portugal. High-high clusters are particularly prominent around Lisbon and Porto, indicating that areas with greater accommodation availability are also those with elevated housing prices. This pattern likely reflects strong market demand in urban centers, where both supply and prices are concentrated. Conversely, in the eastern regions of the country, low-low clusters emerge, pointing to a correlation between limited accommodation supply and lower housing prices. These findings highlight the spatial interdependence between housing availability and market valuation, shaped by broader urbanization and economic dynamics.

6.1 Median bank valuation

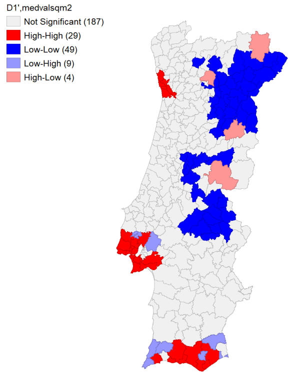

Figure 17: D1

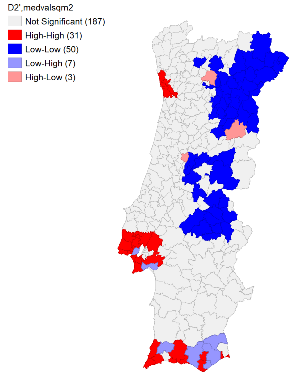

Figure 18: D2

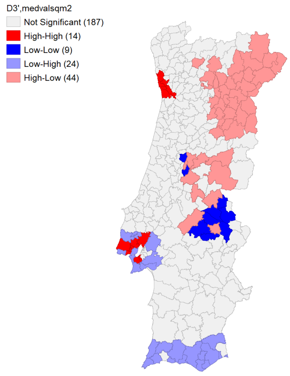

Figure 19: D3

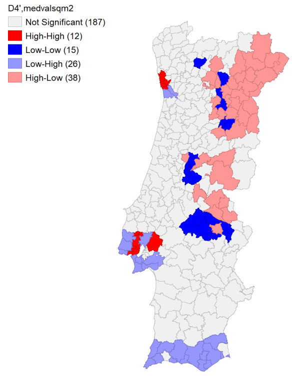

Figure 20: D4

Figure 21: Mapping of spatial autocorrelation, average value of buildings transacted and **demand** principal components

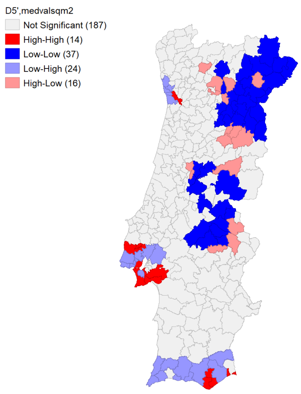

Figure 22: D5

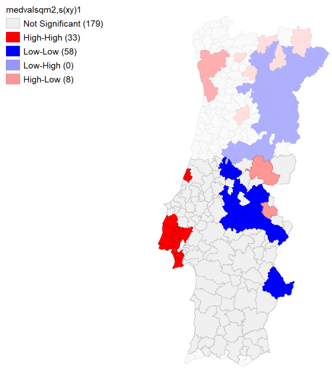

Figure 23: S1

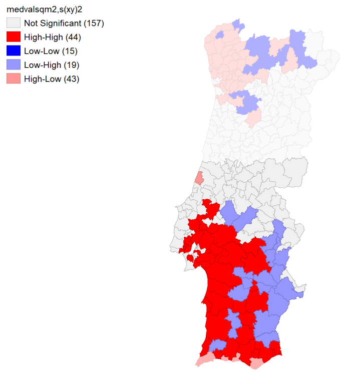

Figure 24: S2

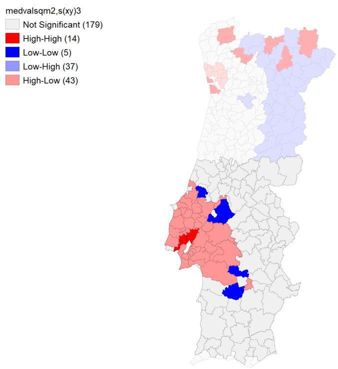

Figure 25: S3

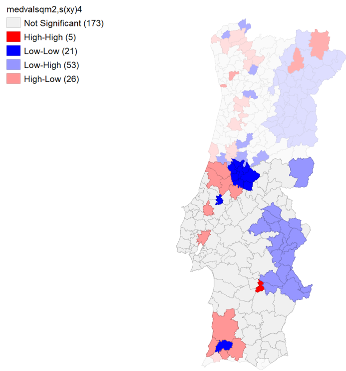

Figure 26: S4

Figure 27: Mapping of spatial autocorrelation, average value of buildings transacted and **supply** principal components

6.2 Average value of buildings transacted

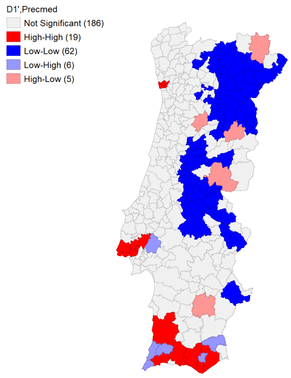

Figure 28: D1

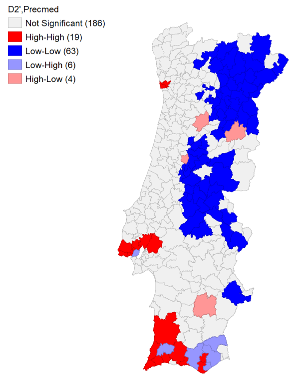

Figure 29: D2

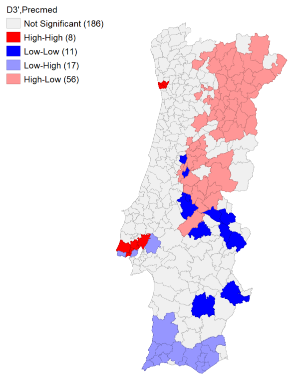

Figure 30: D3

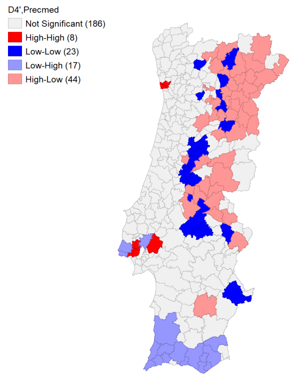

Figure 31: D4

Figure 32: Mapping of spatial autocorrelation, median bank valuation and **demand** principal components

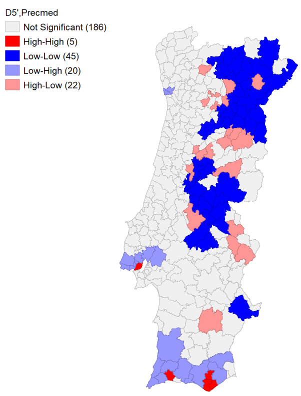

Figure 33: D5

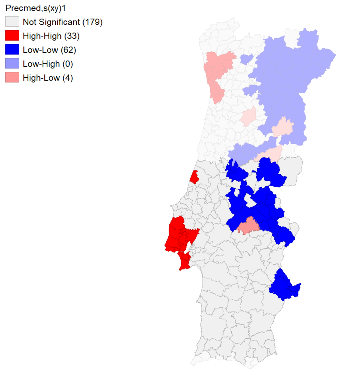

Figure 34: S1

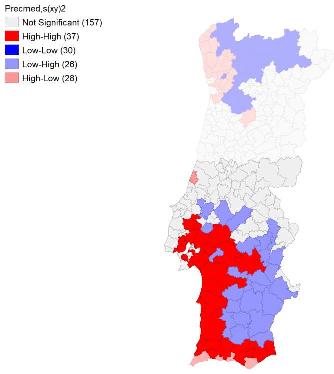

Figure 35: S2

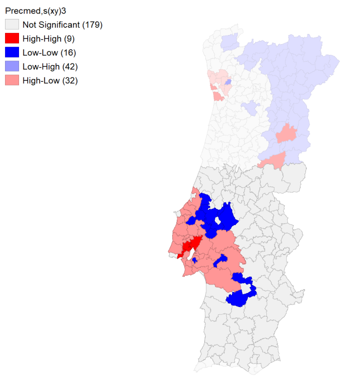

Figure 36: S3

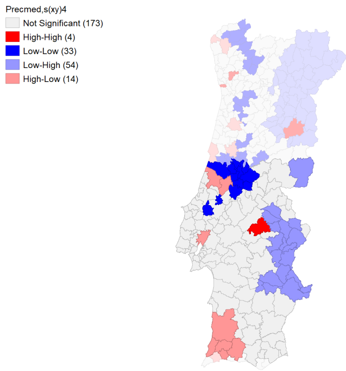

Figure 37: S4

Figure 38: Mapping of spatial autocorrelation, average value of buildings transacted and **supply** principal components

7 OLS of house prices on principal components

After reducing the variable space and defining the principal components, two OLS regressions will be performed. The first one is the 5 demand principal components and 4 supply principal components on the median bank valuation (table 5). The second one is of the 5 demand principal components and 4 supply principal components on the average value of buildings transacted (table 6).

Variable	Coefficient	Std. Error	Probability
Constant	749.77	(15.91)	[0.000]
Demand 1	10.28	(21.33)	[0.630]
Demand 2	102.57	(19.33)	[0.000]
Demand 3	-86.61	(20.68)	[0.000]
Demand 4	-27.77	(16.59)	[0.095]
Demand 5	57.29	(17.03)	[0.001]
Supply 1	130.96	(34.27)	[0.000]
Supply 2	54.71	(27.95)	[0.051]
Supply 3	95.00	(36.21)	[0.009]
Supply 4	-65.02	(17.02)	[0.000]
R-squared	0.751010	F-statistic	89.9203

Table 5: Regression Coefficients for Dep. Var. Median bank valuation

Variable	Coefficient	Std. Error	Probability
Constant	73297.8	(1544.38)	[0.00]
Demand 1	7645.4	(2069.58)	[0.00]
Demand 2	3436.85	(1875.94)	[0.07]
Demand 3	-15765.3	(2007.16)	[0.00]
Demand 4	68.301	(1610.95)	[0.68]
Demand 5	621.526	(1652.28)	[0.71]
Supply 1	3300.6	(3325.51)	[0.32]
Supply 2	-412.055	(2712.58)	[0.88]
Supply 3	4049.5	(3514.36)	[0.25]
Supply 4	-4500.9	(1651.52)	[0.01]
R-squared	0.770692	F-statistic	100.081

Table 6: Regression Coefficients for Dep. Var. Average value of buildings transacted

8 Spatial model selection

As our data comes from the Portuguese municipalities, we have an incentive to test for spatial dependence. A first statistical test to test for this phenomenon is the Moran's I test, which has the following test statistic:

$$I = \frac{n\hat{u}^T W \hat{u}}{\hat{u}^T \hat{u} \left[\sum_i \sum_j w_{ij}\right]}$$

This test statistic is normally distributed (asymptotically). As can be seen, the statistic requires a W matrix to be specified. This W matrix is a spatial weights matrix, which provides information about the contiguity of municipalities. Our W matrix is constructed as a queen matrix, which means that neighbours of a municipality are defined as municipalities that share a common border or vertex with the specific municipality. The diagonal elements are all zero, and the off-diagonal elements are 1 in the case of being a neighbour. Also, the specified W matrix in this research is row-standardized, meaning the test statistic simplifies to:

$$I = \frac{\hat{u}^T W \hat{u}}{\hat{u}^T \hat{u}}$$

In table 7, the result of the Moran's I test is displayed. The null hypothesis of the test is 'no spatial autocorrelation', however a specific alternative hypothesis for this test is not specified. As the test result indicates, the null of no spatial autocorrelation is strongly rejected for both dependent variables, so we accept that there is spatial autocorrelation.

Av. transaction value		*Median bank valuation/m2*	
Test statistic	**P-value**	**Test statistic**	**P-value**
0.5193	$\ll 0.01$	0.1740	0.0001

Table 7: Moran's I test

To find out the specific type of autocorrelation, a Lagrange multiplier test (or Rao score) is executed. The specific type of autocorrelation is crucial information in the model selection stage hereafter, as wrong a inference leads to wrongly specified model. To test for spatial error dependence, the following Lagrange multiplier test is performed:

$$LM_{SEM} = \frac{n^2}{\text{tr}(W^T W + WW)} \left(\frac{\hat{\varepsilon}^T W \hat{\varepsilon}}{\hat{\varepsilon}^T \hat{\varepsilon}}\right)^2 \sim \chi^2(1)$$

Here,

$$H_0 : \lambda = 0$$
$$H_1 : \lambda \neq 0$$

In the model y = X$\beta + u, \quad u = \lambda W u + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 I)$

To test for a spatial lag, the following test is performed:

$$LM_{LAG} = \frac{n^2}{Q} \left(\frac{\hat{\varepsilon}^T W y}{\hat{\varepsilon}^T \hat{\varepsilon}} \right)^2 \sim \chi^2(1)$$

where

$$Q = \left(WX\hat{\beta}\right)^T (I - M_X) \frac{WX\hat{\beta}}{\hat{\sigma}^2} + T, \quad M_X = X \left(X^T X\right)^{-1} X^T, \quad T = \text{tr}(W^T W + WW)$$

Here,

$$H_0 : \rho = 0$$
$$H_1 : \rho \neq 0$$

in the model $y = \rho W y + X\beta + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 I)$

In table 8, the test statistics and respective p-values are reported for the spatial error test and spatial lag test. For both dependent variables, the test statistics strongly reject the null hypothesis of no spatial error and no spatial lag respectively, indicating we that we should be inclined to accept both a spatial error and a spatial lag.

	Av. transaction value		*Median bank valuation/m²*	
Test	**Test statistic**	**P-value**	**Test statistic**	**P-value**
Spatial error	146.57	$\ll 0.01$	12.849	[0.0003]
Spatial lag	152.5	$\ll 0.01$	11.261	[0.0008]

Table 8: Lagrange multiplier tests for spatial dependence

However, these tests both do not control for possible presence of the other. This is why we have to use adjusted (robust) test statistics that control for that. Therefore, the following tests are executed for the detection of a possible spatial error and a spatial lag, respectively:

$$RLM_{SEM} = \frac{1}{T(1 - TQ)} \left[\frac{n\hat{\varepsilon}^T W \hat{\varepsilon}}{\hat{\varepsilon}^T \hat{\varepsilon}} - TQ^{-1} \frac{n\hat{\varepsilon}^T W y}{\hat{\varepsilon}^T \hat{\varepsilon}} \right]^2 \sim \chi^2(1)$$

Here,

$$H_0 : \lambda = 0 \quad | \rho = 0$$
$$H_1 : \lambda \neq 0 \quad | \rho = 0$$

$$RLM_{LAG} = \frac{1}{Q - T} \left[\frac{n\hat{\varepsilon}^T W \hat{\varepsilon}}{\hat{\varepsilon}^T \hat{\varepsilon}} - \frac{n\hat{\varepsilon}^T W y}{\hat{\varepsilon}^T \hat{\varepsilon}} \right]^2 \sim \chi^2(1)$$

Here,

$$H_0 : \rho = 0 \quad | \lambda = 0$$
$$H_1 : \rho \neq 0 \quad | \lambda = 0$$

The robust tests account for the potential presence of both spatial error and spatial lag simultaneously, allowing us to find the source of spatial dependence. Interestingly, now it can be observed from table 8 that for the average transaction value, only the spatial lag is statistically significant. Although the SARAR test (which jointly captures spatial lag and error effects) turns out to be significant, we already concluded that the spatial error is statistically insignificant. Ultimately, we conclude that there is spatial autocorrelation in the form of a *spatial lag*. For the median bank valuation, the spatial error and the spatial lag become insignificant individually when controlled for by the other. However, the SARAR is significant.

	Av. transaction value		*Median bank valuation/m²*	
Test	**Test statistic**	**P-value**	**Test statistic**	**P-value**
Adjusted spatial error	0.9335	[0.334]	1.7838	[0.1817]
Adjusted spatial lag	6.8589	[0.00882]	0.1954	[0.6585]
SARAR	153.43	$\ll 0.01$	13.045	[0.0015]

Table 9: Lagrange multiplier tests for spatial dependence

Concluding, for the average transaction value, a model with a spatial lag will be estimated. This is somewhat logical as the spatial correlation was positive (table 7), and more expensive houses tend to be in neighbourhoods with other expensive houses and the otherway around. For the median bank valuation, a SARAR will be estimated. This is because the individual presence for a spatial lag or a spatial error is both rejected (when controlled for by the other).

Next, an Ansalin test will be conducted for the possible presence of heteroscedasticity. The test statistic and the null and alternative hypotheses are the following:

$$LM = \frac{1}{2}\mathbf{F}^T\mathbf{Z}(\mathbf{Z}^T\mathbf{Z})^{-1}\mathbf{Z}^T\mathbf{F} + \frac{n^2}{Q}\left[\frac{\hat{\varepsilon}^T\mathbf{W}\hat{\varepsilon}}{\hat{\varepsilon}^T\hat{\varepsilon}}\right] \sim \chi^2(p+1)$$

$$H_0 : \rho = 0,\ Var(\varepsilon \mid \mathbf{Z}) = \sigma_\varepsilon^2\mathbf{I}_n \qquad H_1 : \rho \neq 0,\ Var(\varepsilon \mid \mathbf{Z}) = \mathbf{\Sigma}_n$$

In table 10, the test results are visualised. Null of constant variance (Heteroskedasticity) Reject null -> Need robust standard errors As can be concluded, there is heteroscedasticity in the error terms. Hence, we need to control for that when estimating a model.

Test statistic	**P-value**
156.5041	0.000

Table 10: Anselin test for heteroscedasticity

9 Model estimation

Next , the models for both dependent variables will be estimated. In table 10, the regression output is displayed. For the average transaction value a spatial lag model with White rubust standard errors of the following form is used:

$$\mathbf{y} = \rho \mathbf{W}\mathbf{y} + \mathbf{X}\boldsymbol{\beta} + \varepsilon$$

Here, $|\rho|$ is smaller than 1 and X is a vector of the 9 principal components. For the median bank valuation variable, a SARAR model with robust standard errors of the following form is estimated:

$$\mathbf{y} = \rho \mathbf{W}\mathbf{y} + \mathbf{X}\boldsymbol{\beta} + \mathbf{u}$$

$$\mathbf{u} = \lambda \mathbf{W}\mathbf{u} + \varepsilon$$

$$\varepsilon \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$$

Again, $|\rho|$ is smaller than 1 and $|\lambda|$ is smaller than 1.

Table 11: Regression output

	Av. transaction value			*Median bank valuation/m²*		
PC	**Coefficient**	**Standard error**	**P-value**	**Coefficient**	**Standard error**	**P-value**
Intercept	18449.23	12457.75	0.14	156.79	100.04	0.1171
D1	271.40	3220.18	0.93	1.46	20.34	0.94
D2	-1594.09	2546.09	0.53	2.03	24.00	0.93
D3	836.03	2755.45	0.76	-1.70	22.39	0.94
D4	5181.61	2806.77	0.06	32.93	26.06	0.21
D5	329.41	2681.05	0.90	-8.37	20.06	0.68
S1	1038.17	4802.35	0.83	21.06	33.50	0.53
S2	648.69	3610.57	0.86	-8.83	32.33	0.78
S3	-3678.82	5458.22	0.50	16.48	39.76	0.68
S4	-787.79	1730.47	0.65	29.97	20.94	0.15
ρ	0.732	0.16455	$\ll 0.01$	0.85	0.11	$\ll 0.01$
λ				-0.74	0.16	$\ll 0.01$

PC	Direct	Indirect	Total
D1	319.51	693.20	1012.71
D2	-1876.60	-4071.45	-5948.06
D3	984.20	2135.30	3119.50
D4	6099.91	13234.26	19334.17
D5	387.79	841.35	1229.14
S1	1222.15	2651.56	3873.71
S2	763.65	1656.81	2420.47
S3	-4330.78	-9395.99	-13726.78
S4	-927.40	-2012.079	-2939.48

Table 12: Partial effect matrix, av. value of buildings transacted

PC	Direct	Indirect	Total
D1	0.006039	0.013103	0.019142
D2	-0.035484	-0.076983	-0.112467
D3	0.018610	0.040360	0.058970
D4	0.115310	0.250139	0.365449
D5	0.007332	0.015909	0.023241
S1	0.023110	0.050141	0.073251
S2	0.014439	0.031320	0.045759
S3	-0.081898	-0.177639	-0.259537
S4	-0.017537	-0.038033	-0.055570

Table 13: Standardized partial effects matrix, av. value of buildings transacted

PC	Direct	Indirect	Total
D1	1.97	7.83	9.79
D2	2.74	10.90	13.63
D3	-2.28	-9.09	-11.38
D4	44.29	176.42	220.72
D5	-11.25	-44.84	-56.10
S1	28.32	112.82	141.15
S2	-11.88	-47.32	-59.21
S3	22.17	88.30	110.47
S4	40.30	160.52	200.82

Table 14: Partial effects matrix, Median bank valuation/m2

PC	Direct	Indirect	Total
D1	0.005131	0.020392	0.025523
D2	0.007134	0.028383	0.035517
D3	-0.005937	-0.023667	-0.029605
D4	0.115341	0.459567	0.574908
D5	-0.029307	-0.116811	-0.146118
S1	0.073744	0.293889	0.367633
S2	-0.030944	-0.123286	-0.154231
S3	0.057748	0.229985	0.287733
S4	0.104963	0.417934	0.522896

Table 15: Standardized partial effects matrix, Median bank valuation/m2

The spatial regression analysis reveals highly significant values for both the spatial autocorrelation parameter (rho) and the spatial error coefficient (lambda), indicating a strong spatial dependence in housing prices. This suggests that space itself plays a substantial explanatory role in understanding price distributions, particularly at a broader scale. A notable pattern emerges in which high-priced neighborhoods are frequently adjacent to other high-priced areas, reinforcing the spatial clustering of wealth. Despite including a range of property-level features, their estimated marginal effects on prices appear relatively modest when compared to the overarching influence of spatial location. Both models estimate direct, indirect, and total impacts of property-specific features on the two dependent variables, these effects are relatively small when compared to the standard deviation of each respective dependent variable. This indicates that individual property characteristics explain only a limited portion of the variation in housing outcomes. As a result, any substantial reduction in housing prices would likely necessitate broad and aggressive policy interventions according to our models. However, the potential impacts of such policies remain speculative, as they fall outside the range of conditions observed in the available data.

Interestingly, some findings run counter to our initial expectations. Notably, the tourism-related variable does not exhibit a statistically significant effect on housing prices, despite the intuitive assumption that areas with higher tourism activity would experience upward pressure on property values. Similarly, in-country migration, which is commonly believed to influence housing demand, particularly in more attractive or economically vibrant regions, also shows no significant impact in the model.

10 Discussion

Our spatial regression models suggest strong special dependence and the importance of space as a predictor for real estate prices. However other features we assumed to be important predictures as well based on Local morans I analyses do not have the influence we expect in our model. That's why we want to discuss slightly different modeling modifications which could be done in further analyses of portuguese real estate market dynamics.

We propose enhancing the modeling framework to better capture regional heterogeneity and complex interdependencies. Given the pronounced differences across Portuguese regions in terms of real estate market dynamics, a clustering approach could allow for the estimation of cluster-specific models that reflect specific drivers of housing demand and supply to different parts of Portugal. Furthermore, we suggest that key variables—particularly those potentially central to price formation—should be excluded from dimensionality reduction techniques such as principal component analysis, as their independent effects could be masked. Additional explanatory variables should also be incorporated, including regional wage levels.